Día_2

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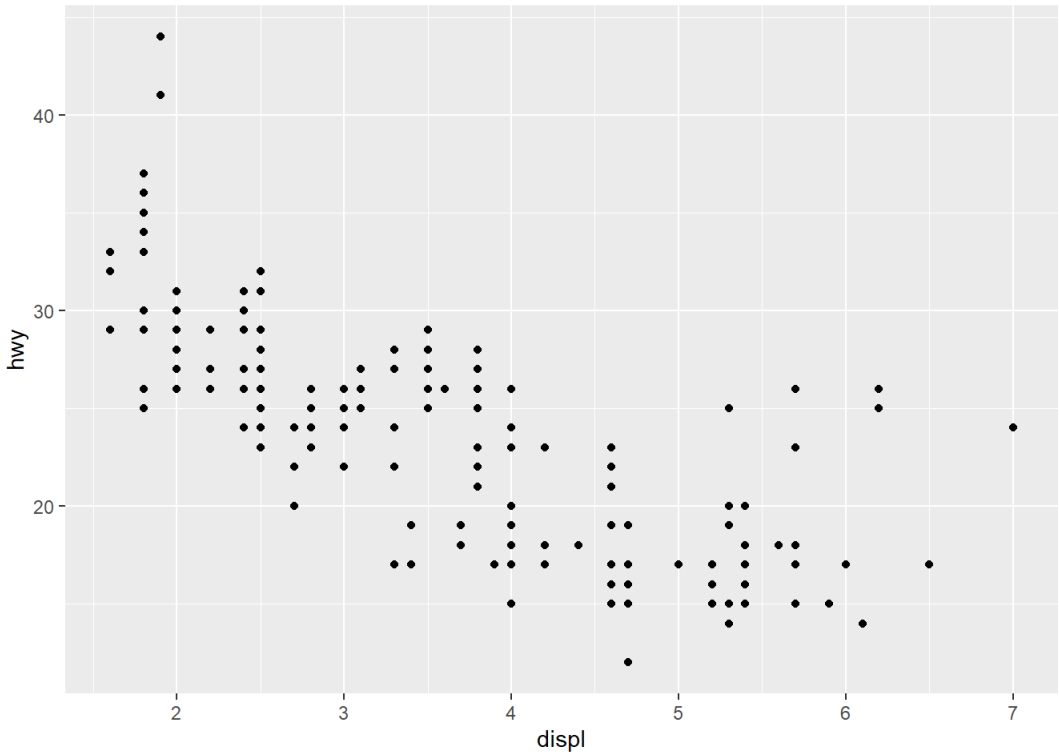
```
library(ggplot2)
library(dplyr)
```

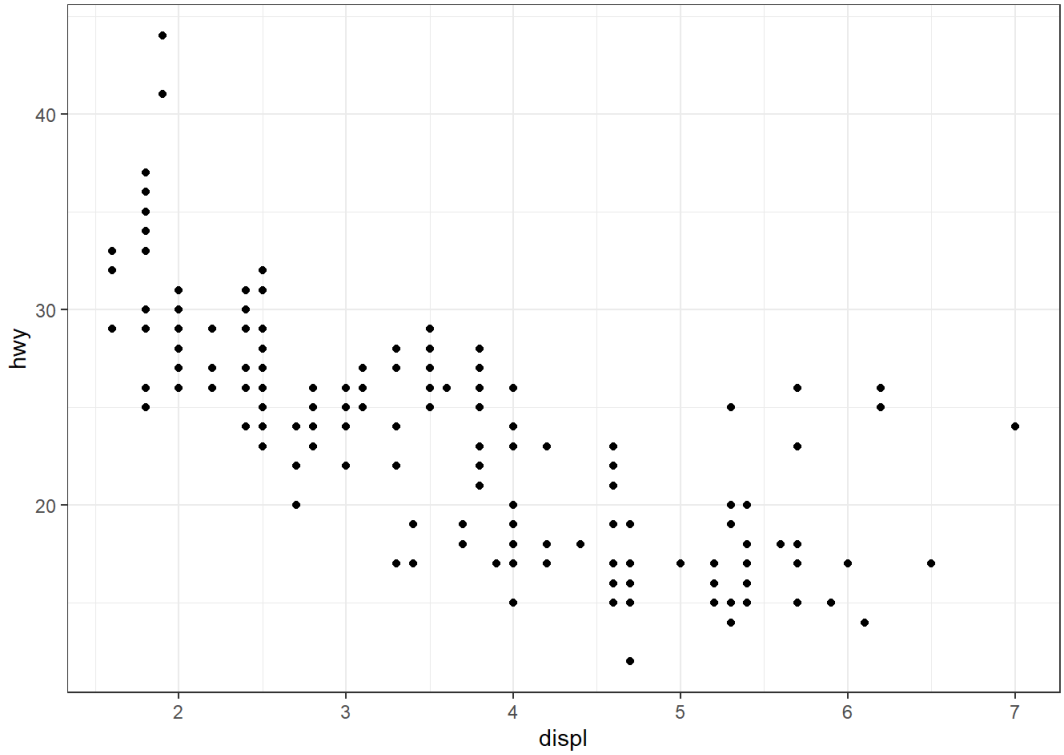
EJEMPLOS GRÁFICAS

scatter plot:

```
Fig1 <- ggplot(data = mpg) + geom_point(aes(x = displ, y = hwy))
Fig1
```

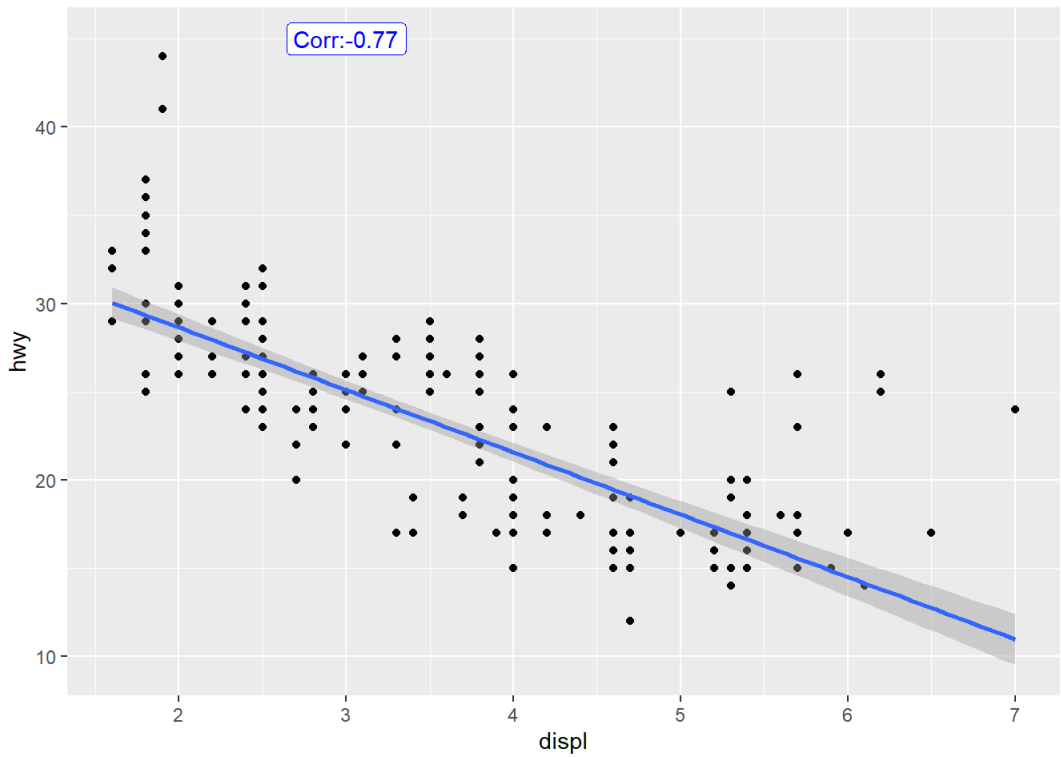



```
Fig1+theme_bw()
```



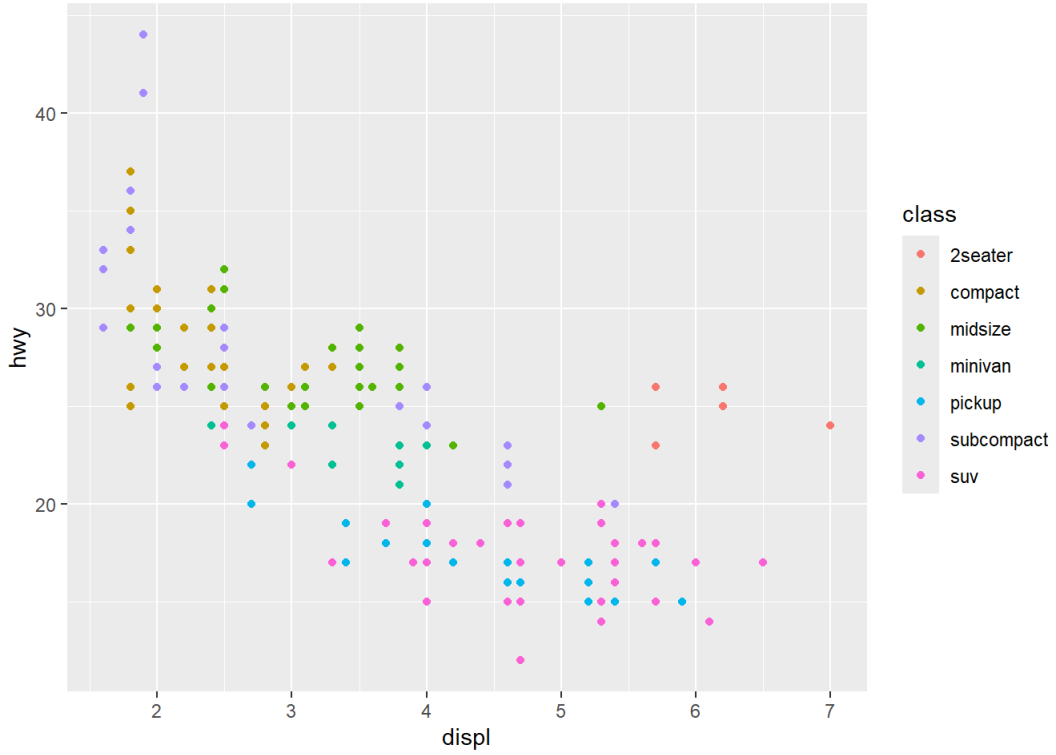
Example scatter plot with regression line

```
Fig2 <- Fig1 +  geom_smooth(aes(x = displ, y = hwy),
                            method = 'lm', formula = y ~ x)
c_h <- round(cor(mpg$displ, mpg$hwy, method = 'pearson'),2)
Fig2 <- Fig2 + geom_label(aes(x = 3, y = 45),
                          label = paste0('Corr:',as.character(c_h)),
                          show.legend = F, fill = 'white', color = 'blue')
Fig2
```



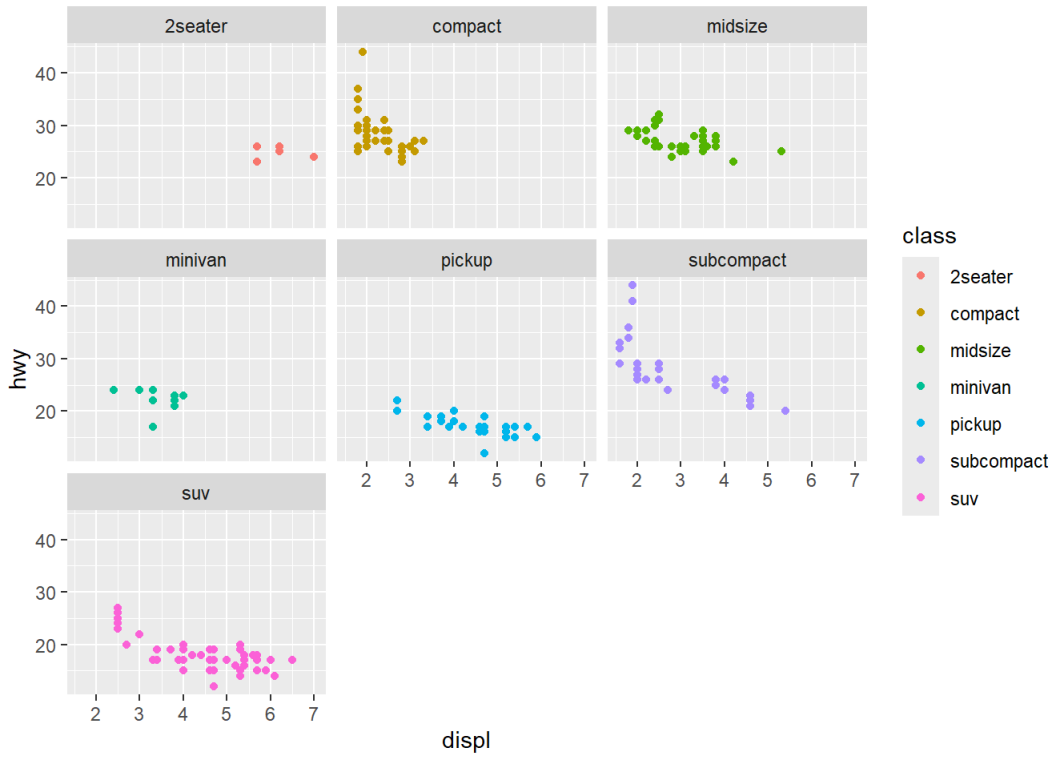
Example scatter plot with color

```
Fig3 <- ggplot(data = mpg) + geom_point(aes(x = displ,
                                            y = hwy, color= class))

Fig3
```



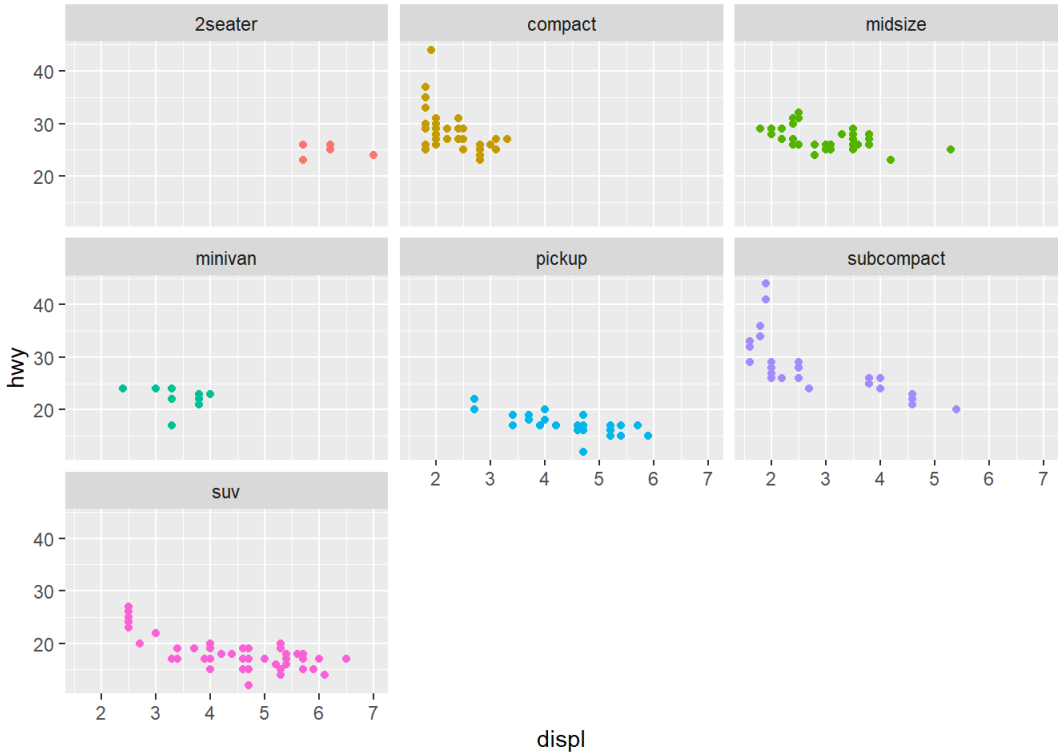
Example scatter plot by facets

```
Fig4 <- Fig3 + facet_wrap(~ class, nrow = 3)
Fig4
```



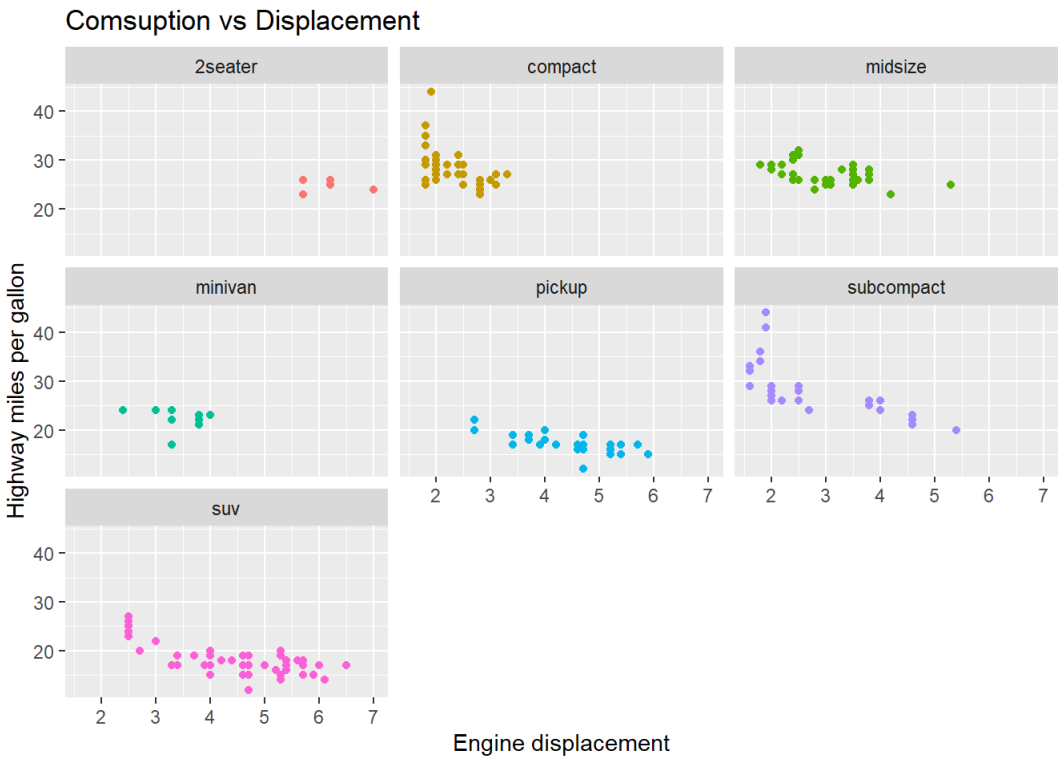
Example scatter plot by facets – remove legend

```
Fig5 <- Fig4 + theme(legend.position = 'none')
Fig5
```



Example scatter plot by facets – change the name of the axes and title

```
Fig6 <- Fig5+
  xlab('Engine displacement') + ylab('Highway miles per gallon') +
  ggtitle ('Comsuption vs Displacement')

Fig6
```

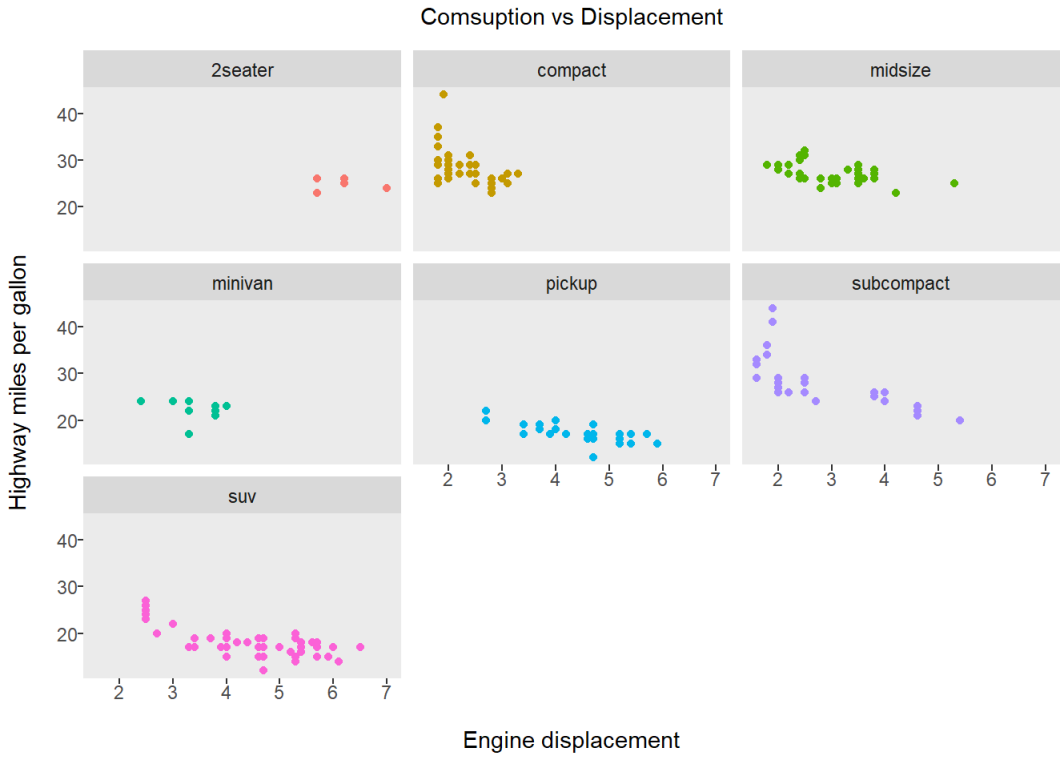


Example scatter plot by facets – change the margin of the text

```
Fig7 <- Fig6 + theme(plot.title = element_text(hjust = 0.5,
                                               margin=margin(0,0,10,0),
                                               size = 11),
                     axis.text.x = element_text(margin=margin(0,0,10,0)),
                     axis.text.y = element_text(margin=margin(0,0,0,10)))

Fig7
```

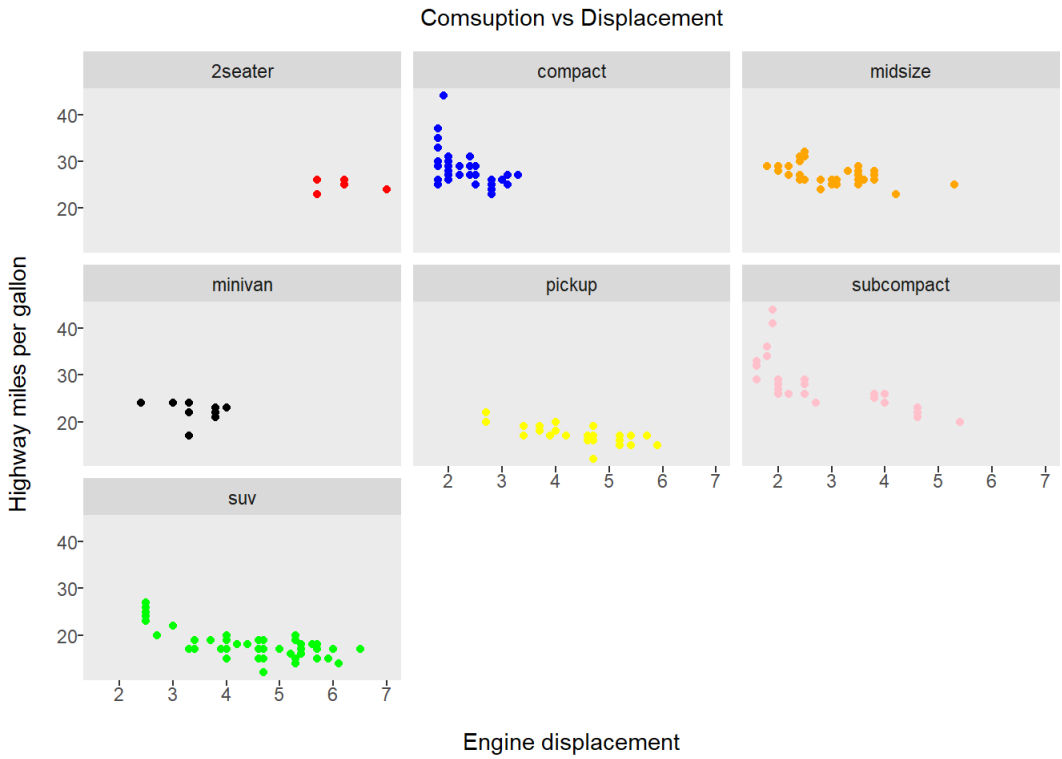
Example scatter plot by facets – change style of the grid

```
Fig8 <- Fig7 + theme(panel.grid.major = element_blank(),
                     panel.grid.minor = element_blank())

Fig8
```



Example scatter plot by facets – change the colors manually and save picture

```
Fig9 <- Fig8 + scale_color_manual(values = c('red', 'blue', 'orange',
                                             'black', 'yellow',
                                             'pink', 'green'))
Fig9
```

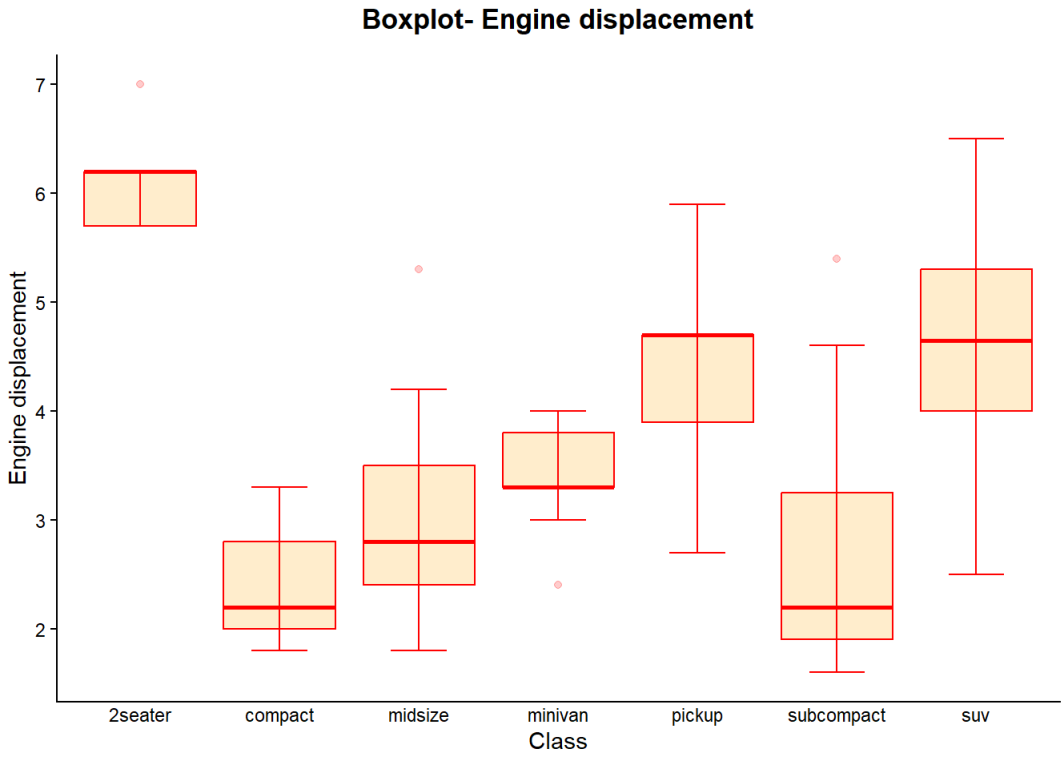



```
#para guardar la figura
# ggsave(filename = paste0(file.path('output','test.png')), height = 10,
#        width = 10 , units = "cm", dpi = 500)
```

##boxplot ########

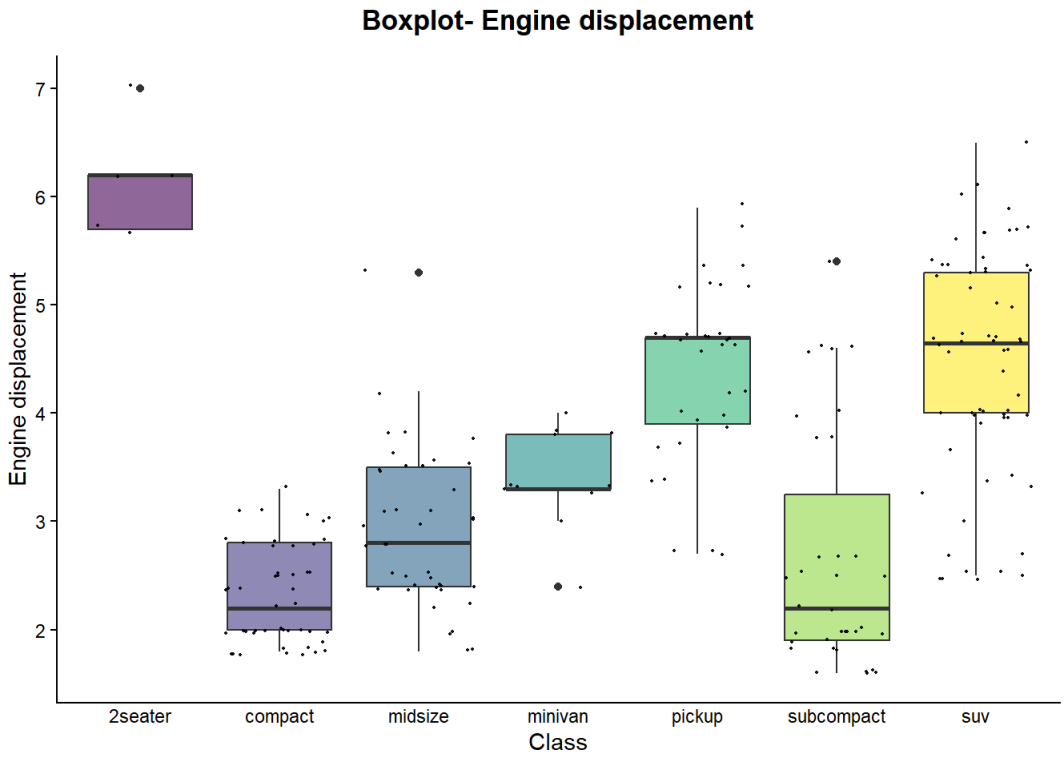
Box-plot of engine-displacement by class of cars

```
Fig10 <- ggplot(data = mpg, aes(class,displ)) +
  geom_boxplot(color="red", fill="orange",
               alpha=0.2, width = 0.8) +
  stat_boxplot(geom = 'errorbar', color="red", width = 0.4)+
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5,
                                  margin=margin(0,0,10,0),
                                  face = 'bold'),
        axis.ticks.x = element_blank()) +
  xlab('Class') + ylab('Engine displacement') +
  ggtitle('Boxplot- Engine displacement')

Fig10
```



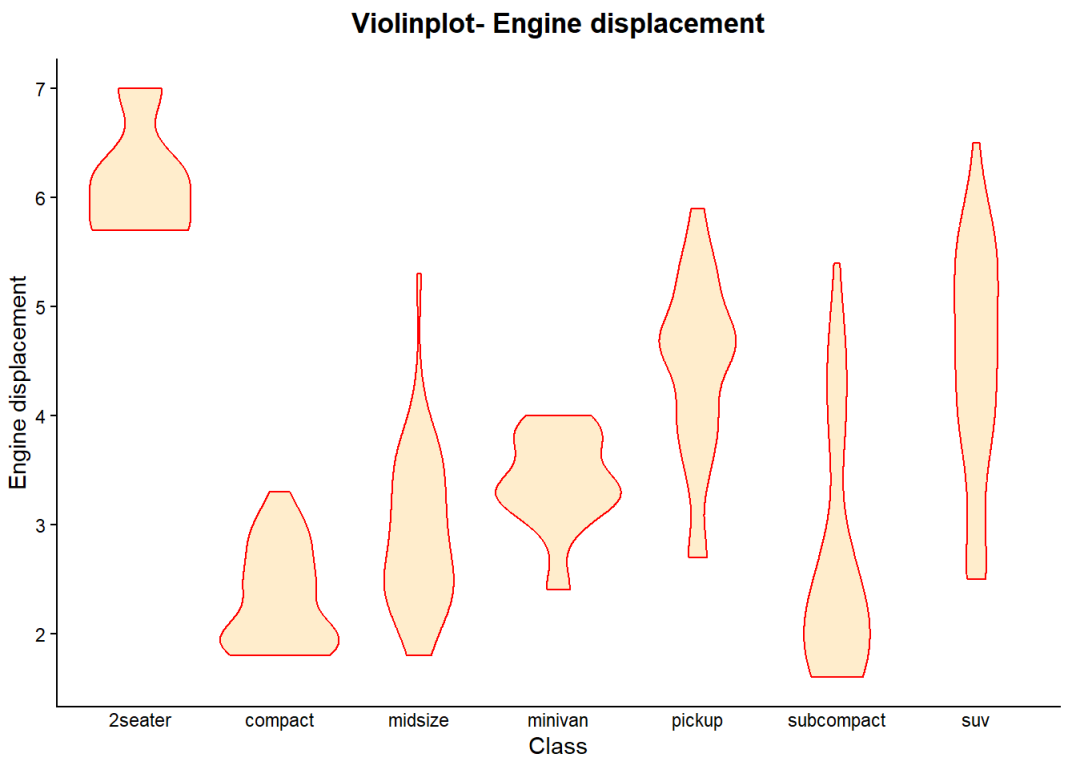
Box-plot of engine-displacement by class of cars

```
library(viridis)
Fig11 <- ggplot(data = mpg, aes(x=class,y=displ,fill=class)) +
  geom_boxplot() +
  scale_fill_viridis(discrete = TRUE, alpha=0.6) +
  geom_jitter(color="black", size=0.4, alpha=0.9) + theme_classic() +
  theme(plot.title = element_text(hjust = 0.5, margin=margin(0,0,10,0),
                                  face = 'bold'), axis.ticks.x = element_blank(),   legend.position="none
") +
  xlab('Class') + ylab('Engine displacement') + ggtitle('Boxplot- Engine displacement')
Fig11
```



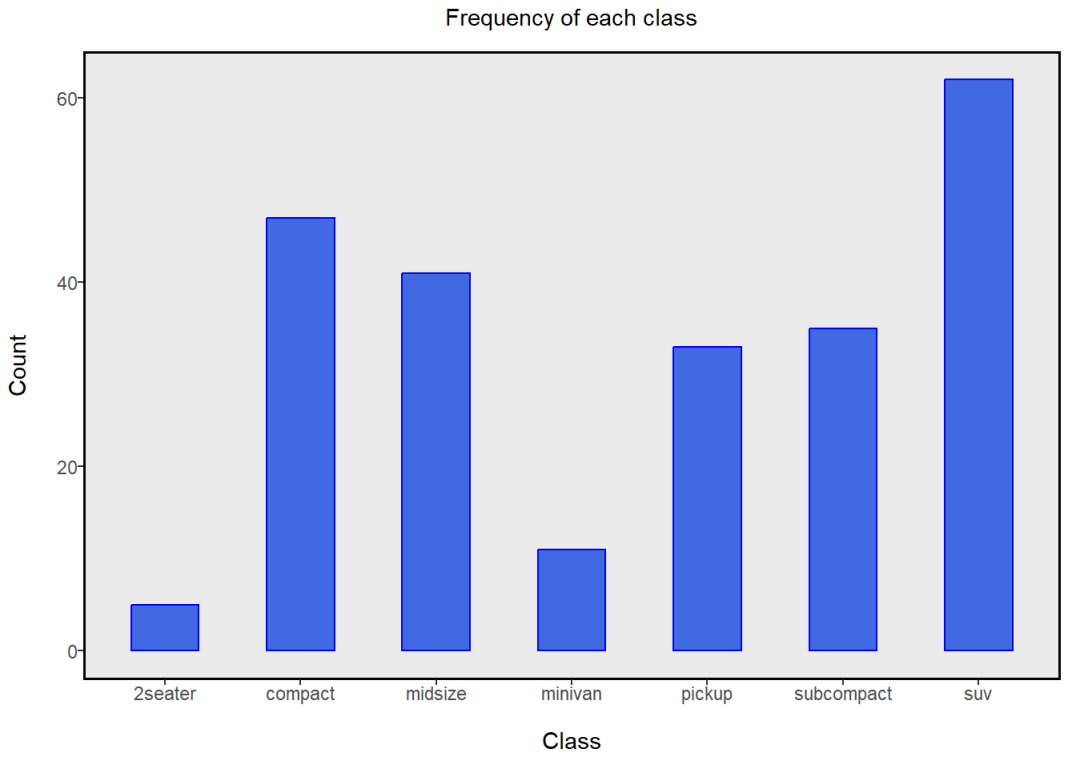
violin plot

```
Fig12 <- ggplot(data = mpg, aes(class,displ)) +
  geom_violin(color="red", fill="orange", alpha=0.2)+ theme_classic() +
  theme(plot.title = element_text(hjust = 0.5, margin=margin(0,0,10,0), face = 'bold'),
        axis.ticks.x = element_blank()) +
  xlab('Class') + ylab('Engine displacement') + ggtitle('Violinplot- Engine displacement')

Fig12
```

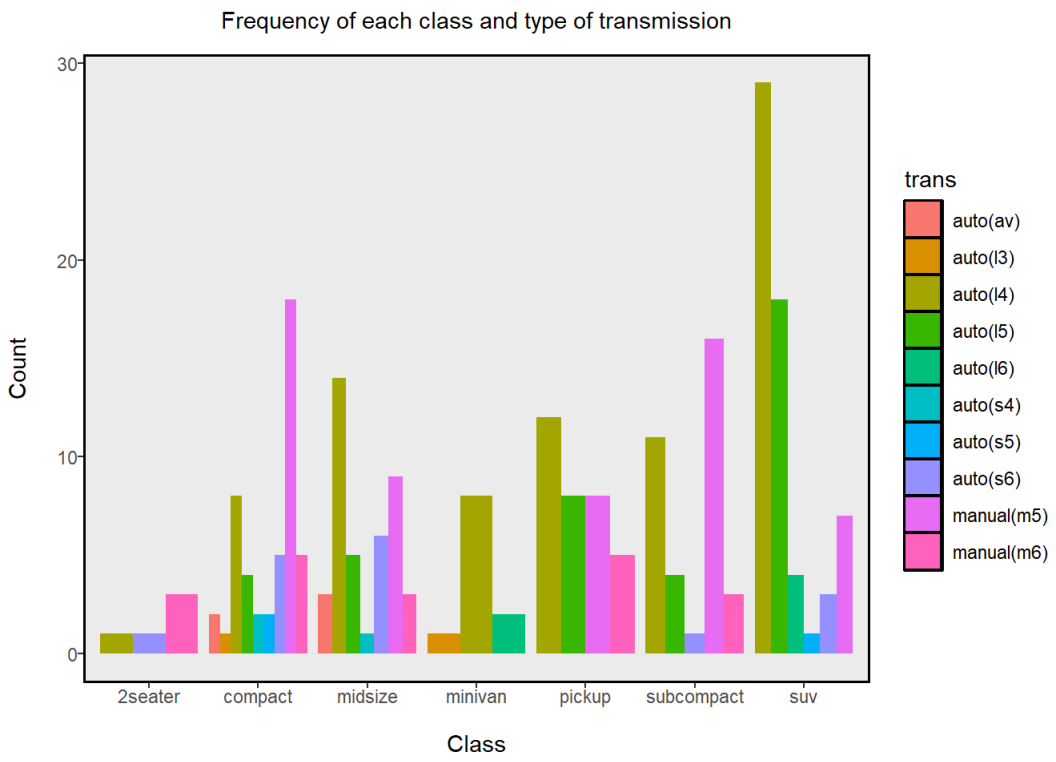


barplot

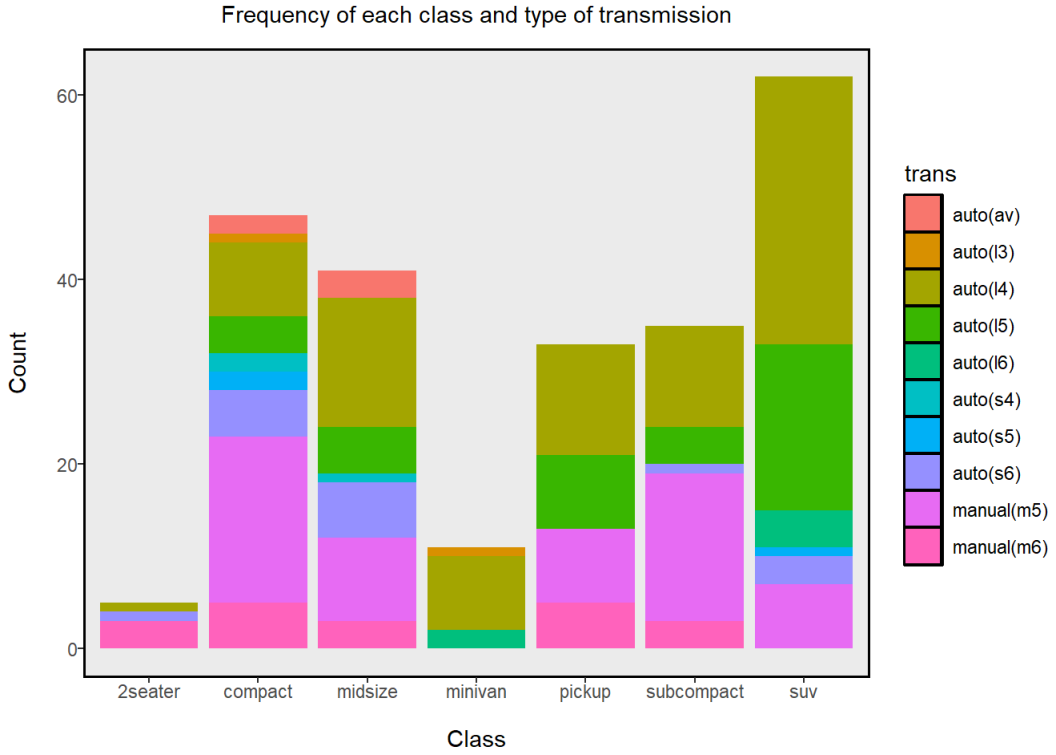
```
Fig14 <- ggplot(data = mpg) + geom_bar(mapping = aes(x = class), width = 0.5,
                                       color = 'blue', fill='royalblue') +
  theme(plot.title = element_text(hjust = 0.5,margin=margin(0,0,10,0), size = 11),
        axis.text.x = element_text(margin=margin(0,0,10,0)), axis.text.y = element_text(margin=margin(0,0
,0,10)), panel.grid.major = element_blank(), panel.grid.minor = element_blank(), panel.background = eleme
nt_rect(colour = "black", size = 1)) +
  xlab('Class') + ylab('Count') + ggtitle('Frequency of each class')

Fig14
```



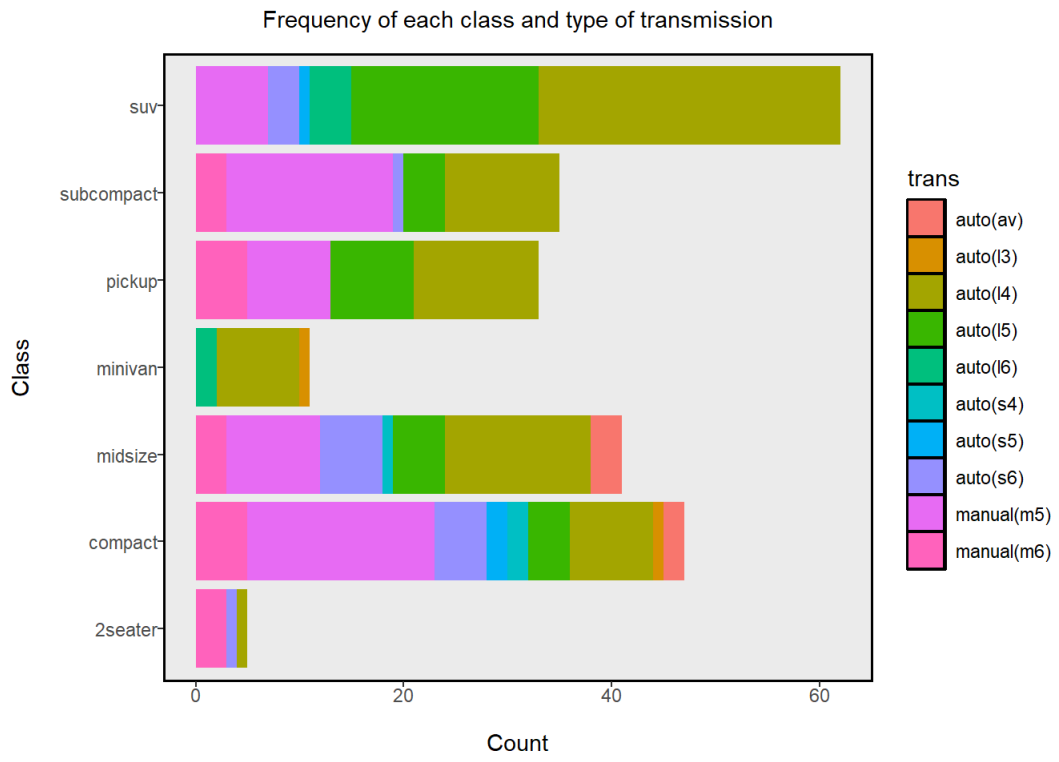

```
Fig15 <- ggplot(data = mpg) + geom_bar(mapping = aes(x = class, fill = trans),
                                       position = 'dodge' ) +
  theme(plot.title = element_text(hjust = 0.5,margin=margin(0,0,10,0), size = 11),
        axis.text.x = element_text(margin=margin(0,0,10,0)), axis.text.y = element_text(margin=margin(0,0
,0,10)), panel.grid.major = element_blank(), panel.grid.minor = element_blank(), panel.background = eleme
nt_rect(colour = "black", size = 1)) +
  xlab('Class') + ylab('Count') + ggtitle('Frequency of each class and type of transmission')
Fig15
```



```
Fig16 <- ggplot(data = mpg) +
  geom_bar(mapping = aes(x = class, fill = trans), position = 'stack' ) +
  theme(plot.title = element_text(hjust = 0.5,margin=margin(0,0,10,0), size = 11),
        axis.text.x = element_text(margin=margin(0,0,10,0)), axis.text.y = element_text(margin=margin(0,0
,0,10)), panel.grid.major = element_blank(), panel.grid.minor = element_blank(), panel.background = eleme
nt_rect(colour = "black", size = 1)) +
  xlab('Class') + ylab('Count') + ggtitle('Frequency of each class and type of transmission')
Fig16
```




```
Fig17 <- ggplot(data = mpg) + geom_bar(mapping = aes(x = class, fill = trans),
                                       position = 'stack' ) + coord_flip() +
  theme(plot.title = element_text(hjust = 0.5,margin=margin(0,0,10,0), size = 11),
        axis.text.x = element_text(margin=margin(0,0,10,0)), axis.text.y = element_text(margin=margin(0,0
,0,10)), panel.grid.major = element_blank(), panel.grid.minor = element_blank(), panel.background = eleme
nt_rect(colour = "black", size = 1)) +
  xlab('Class') + ylab('Count') + ggtitle('Frequency of each class and type of transmission')
Fig17
```




```
mpg$year <- factor(mpg$year, levels= c('1999', '2008'))
Fig18 <- ggplot(data = mpg)+ geom_tile(aes(x = year, y = manufacturer, fill = hwy),colour='black') +  the
me_minimal() +
  theme(axis.title.x=element_blank(),axis.title.y=element_blank(),
        axis.text.x = element_text(hjust = 0.9, vjust = 0.9, angle = 45)) +
  scale_fill_gradient(low="red", high="blue")
Fig18
```

Ejercicio 1

```r
datos<- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Juan/Día 2/data/season-1819_csv
.csv", header = T, sep = ",")

construir_tabla <- function(datos,equipo) {
  win_home <- datos %>%
    filter(HomeTeam == equipo & FTR == "H")
  win_home <- nrow(win_home)
  no_win_home <- datos %>%
    filter(HomeTeam == equipo & FTR != "H")
  no_win_home <- nrow(no_win_home)
  win_away <- datos %>%
    filter(AwayTeam == equipo & FTR == "A")
  win_away <- nrow(win_away)
  no_win_away <- datos %>%
    filter(AwayTeam == equipo & FTR != "A")
  no_win_away <- nrow(no_win_away)

  tabla <- matrix(nrow = 2, ncol = 2)
  tabla[1,1] <- win_home
  tabla[1,2] <- win_away
  tabla[2,1] <- no_win_home
  tabla[2,2] <- no_win_away

  colnames(tabla) <- c("Home","Away")
  rownames(tabla) <- c("Win", "No Win")

  return(tabla)
}

tabla_real <- construir_tabla(datos,"Sociedad")
tabla_real
```

```
##        Home Away
## Win       7    6
## No Win   12   13
```

```r
fisher.test(tabla_real)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tabla_real
## p-value = 1
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.2702689 6.0075392
## sample estimates:
## odds ratio
##   1.256103
```

```r
tabla_barsa <- construir_tabla(datos,"Barcelona")
tabla_barsa
```

```
##        Home Away
## Win      15   11
## No Win    4    8
```

```r
fisher.test(tabla_barsa)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tabla_barsa
## p-value = 0.2953
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   0.5397459 15.3302173
## sample estimates:
## odds ratio
##   2.654629
```

```
cat("En resumen, ni el Barça ni la Real Sociedad dependen de estar en casa para ganar")
```

```
## En resumen, ni el Barça ni la Real Sociedad dependen de estar en casa para ganar
```

Ejercicio 2

```
data <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Juan/Día 2/data/crimeUS.csv", h
eader = T, sep = ",")

data1 <- data %>%
  aggregate(cbind(afam, cauc) ~ state, FUN = mean)

data1$afam <- round(data1$afam)
data1$cauc <- round(data1$cauc)


data2 <- data %>%
  group_by(state) %>%
  summarise(African_American = round(mean(afam)),
            Caucasic = round(mean(cauc)))

head(data2)
```

```
## # A tibble: 6 × 3
##   state      African_American Caucasic
##   <chr>                 <dbl>    <dbl>
## 1 Alabama                   9       55
## 2 Alaska                    8       61
## 3 Arizona                   4       65
## 4 Arkansas                  6       60
## 5 California                7       61
## 6 Colorado                  3       71
```

```
head(data1)
```

```
##        state afam cauc
## 1    Alabama    9   55
## 2     Alaska    8   61
## 3    Arizona    4   65
## 4   Arkansas    6   60
## 5 California    7   61
## 6   Colorado    3   71
```

```r
crear_tabla <- function(estado1, estado2, datos) {

  V1 <- datos %>%
    filter(state == estado1) %>%
    pull(cauc) %>%
    sum()
  V2 <- datos %>%
    filter(state == estado1) %>%
  pull(afam) %>%
    sum()
  V3 <- datos %>%
    filter(state == estado2) %>%
  pull(cauc) %>%
    sum()
  V4 <- datos %>%
    filter(state == estado2) %>%
  pull(afam) %>%
    sum()

  tabla <- matrix(nrow = 2, ncol = 2)
  tabla[1,1] <- V1
  tabla[1,2] <- V2
  tabla[2,1] <- V3
  tabla[2,2] <- V4

  colnames(tabla) <- c("cauc","afam")
  rownames(tabla) <- c(estado1, estado2)

  return(tabla)
}

tabla1 <- crear_tabla("District of Columbia","Hawaii",data)
tabla1
```

```
##                          cauc     afam
## District of Columbia 571.9526 549.3898
## Hawaii               601.0371 573.2988
```

```r
fisher.test(tabla1)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tabla1
## p-value = 0.9667
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.8404327 1.1740918
## sample estimates:
## odds ratio
##  0.9933557
```

```r
tabla2 <- crear_tabla("Florida","Hawaii",data)
tabla2
```

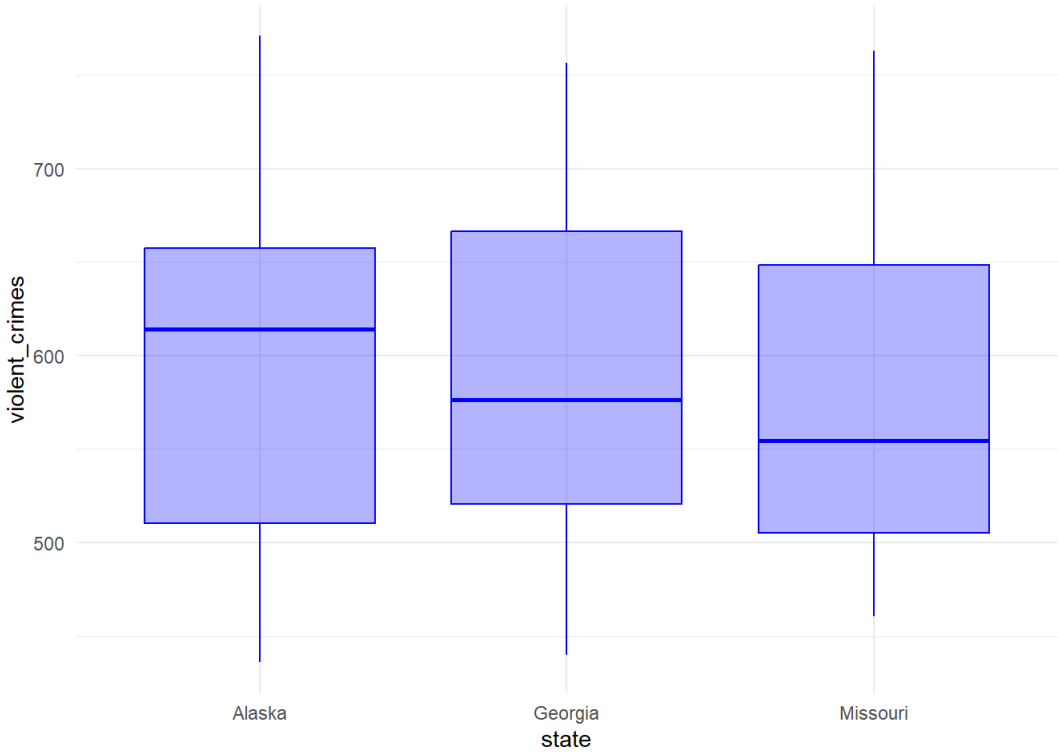
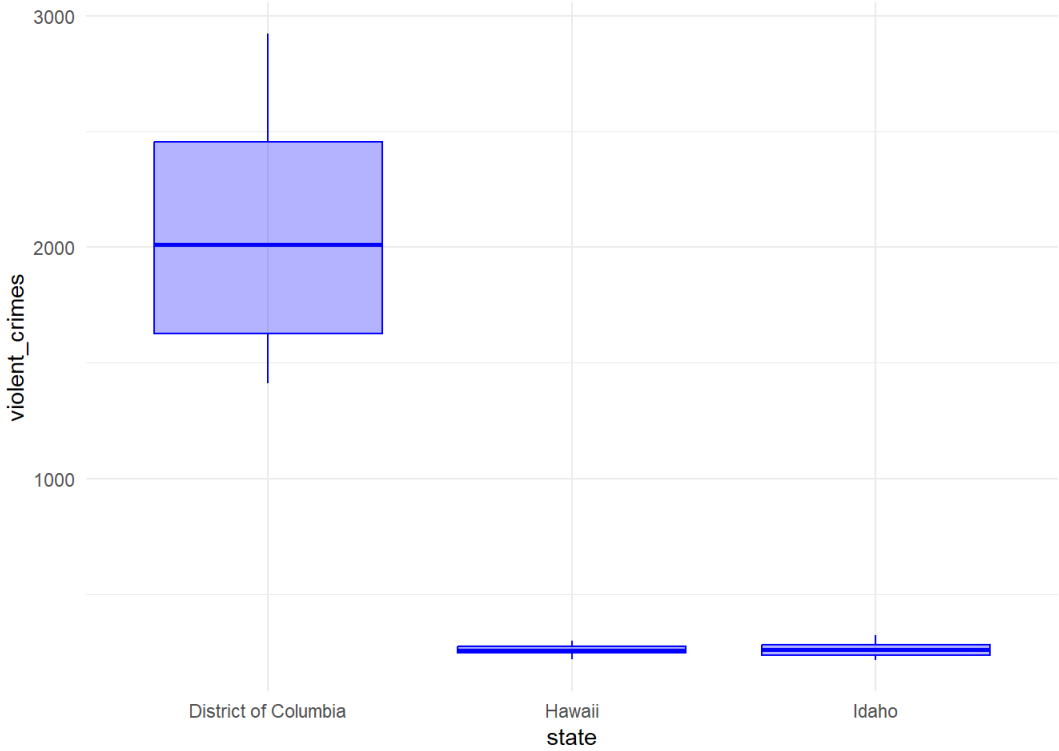
```
##              cauc     afam
## Florida 1349.7396 126.0711
## Hawaii   601.0371 573.2988
```

```r
fisher.test(tabla2)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tabla2
## p-value < 2.2e-16
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   8.199425 12.768945
## sample estimates:
## odds ratio
##   10.20207
```

Ejercicio 3

```
data <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Juan/Día 2/data/crimeUS.csv", h
eader = T, sep = ",")

data1 <- data %>%
  mutate(violent_crimes = (violent/population)*100000)

comparar <- function(s1, s2, s3, data) {
data2 <- data %>%
  group_by(state) %>%
  filter(state %in% c(s1, s2, s3))

modelo <- aov(data2$violent_crimes ~ data2$state)
summary(modelo)

bp <- ggplot(data2, aes(x = state, y = violent_crimes)) +
  geom_boxplot(color = "blue", fill = "blue", alpha = 0.3) +
  theme_minimal()
bp

#return(summary(modelo))
return(bp)
}

comp1 <- comparar("Alaska","Georgia","Missouri", data1)
comp1
```



```
comp2 <- comparar("District of Columbia","Hawaii","Idaho", data1)
comp2
```